

Module – 2

Data pre-processing

By

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Data Pre-Processing

Involves

- Understand the type of data given in the input dataset
- Explore the data to understand its quality
- Explore the relationship among the features (Inter-feature relationship)
- Find potential issues in data
- Remediate data if needed
- Identification of highly relevant features (Feature Selection / Feature Subset Selection)
 - A key to the success of Data Mining lies in the fact that the features are less in number as well as the similarity between each other is very less
- A new set of features are extracted from the original features which are quite dissimilar in nature (Feature Extraction)
 - An 'n' dimensional feature space gets transformed to an 'm' dimensional feature space, where the dimensions are orthogonal to each other, i.e. completely independent of each other
- Augments the feature space by creating additional features (Feature Construction)
 - If there are 'n' features or dimensions in a dataset, after feature construction, 'm' more features or dimensions may get added Hence the dataset will become 'n+m' dimension

AN EXAMPLE

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Name	Program	Date of Birth	Age	Admission year	Pass out year
Abhaya	B. Tech (IT)	01.03.2006	18	2024	2028
Vishal	B. Tech (IT)	05.06.2005	19	2023	2027
Dhruv	B. Tech (IT)	08.08.2005	19	2023	2027
Neel	B. Tech (IT)	07.10.2005	18	2023	2027
Aditya	B. Tech (IT)	03.02.2006	18	2024	2028
Kumar	M. Tech (SE)	06.04.2006	18	2024	2029
Hari	M. Tech (SE)	08.07.2005	19	2023	2028

SOME REDUNDANCIES

- The **column “Age” is redundant** as it can be computed from the date of birth provided and from the current date at any time
- Similarly, the **year of passing** is also redundant as it can be computed by knowing the duration of the course and the year of admission
- So, **we can go ahead and delete these two columns** from the dataset
- This will **reduce the complexity** of the table and decrease the overall time for processing of the model

ROLL MULTIPLE FEATURES INTO ONE

- Let us consider the following table: (PowerAde is the official sports drinks of Australian rugby players)
- The Table contains a list of products sold on an e-commerce platform
- There are 4 buyers and 8 products (In reality there may be several buyers)

-	Protein Shake	Nike Sneakers	Adidas boots	Fitbit	PowerAde	Protein Bar	Fitness Watch	Vitamins
Buyer 1	1	1	0	1	0	5	1	0
Buyer 2	0	0	0	0	0	0	0	1
Buyer 3	3	0	1	0	5	0	0	0
Buyer 4	1	1	0	0	10	1	0	0

MULTIPLE FEATURES ROLLED INTO ONE TABLE

- In order to analyze data in a more efficient way, we can reduce the number of columns by **merging similar features into fewer columns**
- We can replace the products with product sub-types
- We can have **3 product subtype columns**
- **Health Food**(Protein shake, protein Bar, Vitamins, powerade) , **Apparel** (Nike Sneakers, Adidas Boots) and **Digital** (Fitbit, Fitness watch)

	Health Food	Apparel	Digital
Buyer 1	6	1	2
Buyer 2	1	0	0
Buyer 3	8	1	0
Buyer 4	12	1	0

Attributes / Dimensions / Features/Variables

- Types of attributes based on values that the attribute can hold
 - **Continuous**
 - **Discrete**
- **Continuous**
 - A variable where an infinite number of numeric values are possible within a specific range. Eg. Temperature
- **Discrete**
 - A variable that can only take a certain number of values (either text or numbers)
 - Eg. Color where values could be black, blue, red, yellow, and so on, or the variable Score where the variable can only take values 1, 2, 3, 4, or 5

Attributes / Dimensions / Features/Variables

- **Binary**

- A variable where there are only two values
- Eg. Variable **Gender** whose values can be Male or Female
- Eg. variable **Purchase** may indicate whether a customer bought a particular product and the convention that was used to represent the two cases is 0 (did not buy) and 1 (did buy).
- **Symmetric binary**: both outcomes are equally important
 - e.g., Gender
- **Asymmetric binary**: outcomes not equally important.
 - e.g., medical test (positive vs. negative)
 - Convention: assign 1 to most important outcome

Attributes / Dimensions / Features/Variables

- **Types of attributes based on measurement scale**
 - **Nominal**
 - **Ordinal**
 - **Interval**
 - **Ratio**
- **Nominal**
 - It is not possible to determine whether one value is larger than another
 - For example, a variable **Industry** would be nominal where it takes values such as financial, engineering, retail, and so on. The order of these values has no meaning

Attributes / Dimensions / Features/Variables

- **Ordinal**

- Describes a variable whose values are ordered
- However, the difference between the values does not describe the magnitude of the actual difference
- Eg. **Rank** attribute may have the value Low, High, Medium
- It is impossible to determine the magnitude of the difference between the three values

- **Interval**

- Attributes, where the interval between their values has meaning
- Eg. when looking at three data points measured on the Fahrenheit scale, 5 F, 10 F, 15 F, the differences between the values from 5 to 10 and from 10 to 15 are both 5
- Interval scales hold no true zero and can represent values below zero

Attributes / Dimensions / Features/Variables

- **Ratio**

- The ratio of two measurements has a meaningful interpretation
- Eg. **Weight** is a ratio variable, a weight of 100 grams is 10 times as heavy as a weight of 10 grams
- Ratio variables, never fall below zero.

Summary of Measurement

	Meaningful order	Meaningful difference	Natural zero
Nominal	No	No	No
Ordinal	Yes	No	No
Interval	Yes	Yes	No
Ratio	Yes	Yes	Yes

Check the attributes type

Gender	Age	Weight	LoveShopping?
1	36	60	1
0	24	45	5
0	20	40	4
1	25	55	1
0	43	53	3

Check the attributes type

	Outlook	Temperature	Humidity	Windy	PlayGolf
0	sunny	85	85	FALSE	no
1	rainy	80	90	TRUE	no
2	overcast	83	86	FALSE	yes
3	rainy	70	96	FALSE	yes
4	rainy	72	80	FALSE	yes
5	rainy	65	70	TRUE	no
6	overcast	64	65	TRUE	yes
7	sunny	72	95	FALSE	no
8	sunny	69	70	FALSE	yes
9	rainy	75	80	FALSE	yes
10	sunny	75	70	TRUE	yes
11	overcast	72	80	TRUE	yes
12	overcast	81	75	FALSE	yes
13	rainy	71	91	TRUE	no

Various Feature Engineering Tasks

apartment_ length	apartment_ breadth	apartment_ price		apartment_ length	apartment_ breadth	apartment_ area	apartment_ price
80	59	23,60,000		80	59	4,720	23,60,000
54	45	12,15,000		54	45	2,430	12,15,000
78	56	21,84,000	→	78	56	4,368	21,84,000
63	63	19,84,000		63	63	3,969	19,84,500
83	74	30,71,000		83	74	6,142	30,71,000
92	86	39,56,000		92	86	7,912	39,56,000

Feature construction (example 1)

Various Feature Engineering Tasks

Age (Years)	City of origin	Parents athlete	Chance of win
18	City A	Yes	Y
20	City B	No	Y
23	City B	Yes	Y
19	City A	No	N
18	City C	Yes	N
22	City B	Yes	Y

Age (Years)	origin_city_A	origin_city_B	origin_city_C	parents_athlete_Y	win_chance_Y
18	1	0	0	1	1
20	0	1	0	0	1
23	0	1	0	1	1
19	1	0	0	0	0
18	0	0	1	1	0
22	0	1	0	1	1

78

Feature construction (encoding nominal variables)

Various Feature Engineering Tasks

apartment_area	apartment_price
4,720	23,60,000
2,430	12,15,000
4,368	21,84,000
3,969	19,84,500
6,142	30,71,000
7,912	39,56,000

(a)

apartment_area	apartment_grade
4,720	Medium
2,430	Low
4,368	Medium
3,969	Low
6,142	High
7,912	High

(b)

Feature construction (numeric to categorical)

Role of attribute in Data Mining

- **Descriptors**

- Variables which are used as the input in both creating a predictive model and generating predictions from the models
- Also called as **Independent attributes**
- Also called as **X variables**

- **Response variables**

- These variables (usually one variable) are predicted using the descriptor variables as inputs
- Also called as **Dependent Attribute, class label, classifier**
- They are also referred to as **Y variables**

Find the X variables and Y variable

- **A car dataset has the following attributes**
 - Vehicle identification number
 - Manufacturer
 - Weight
 - Number of cylinders
 - Fuel efficiency

Measuring Central Tendency

- **Mean :** The most common and effective numeric measure of the “center” of a set of data is the (arithmetic) mean.
- The mean of a set of data is the sum of the data values divided by the number of values.

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}.$$

Weighted arithmetic mean or the weighted average.

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_N x_N}{w_1 + w_2 + \cdots + w_N}.$$

Measuring Central Tendency

Median :

- For skewed (asymmetric) data, a better measure of the center of data is the median, which is the middle value in a set of ordered data values.
- It is the value that separates the higher half of a data set from the lower half.
- Middle value of the variable once it has been sorted from smallest to largest
- If the number of data values is odd, then the median can be found by $[(n + 1)/2]$ th term

Data Given : 3,4,7,2,3,7,4,2,4,7,4

Sorted Data : 2,2,3,3,4,4,4,4,7,7,7

The 6th term, Median = 4

- If the number of data values is even, then the median can be found by $[(n/2)$ th term + $((n/2) + 1)$ th term]/2

Data Given : 5, 10, 8, 6, 4, 8, 2, 5, 7, 7

Sorted Data : 2, 4, 5, 5, 6, 7, 7, 8, 8, 10

Median : $(6+7)/2 = 6.5$

Measuring Central Tendency

- **Mode** – Mode for a set of data is the value that occurs most frequently compared to all neighboring values in the set

3,4,5,6,7,7,7,8,8,9

Mode = 7

Find the mode of the data set

3,4,5,6,7,7,7,8,8,8,9

Mode = 7.5

- **Mid range** - Can also be used to assess the central tendency of a numeric data set. It is the average of the largest and smallest values in the set

Midrange = 6

Measuring Central Tendency

Mode for categorical data

Vehicle color survey data is provided here. Which color is the mode?

Color	Frequency
Blue	3
Green	5
Red	4
White	3
Black	2
Grey	3

It is possible for a data set to have more than one mode if several categories have the same frequency, or no modes if each every category occurs only once

Measuring the dispersion of Data

- Range

- Measure the “spread” of the data
- Simple measure of the variation for a particular variable. It is calculated as the difference between the highest and lowest values.

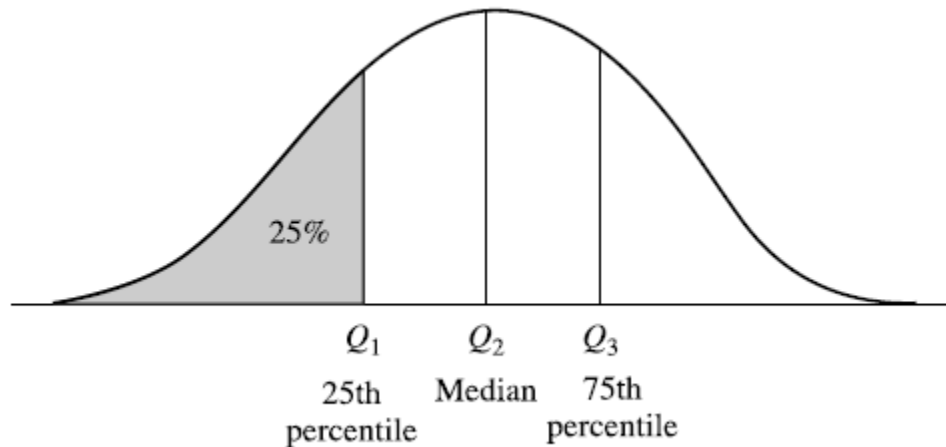
2,3,4,6,7,7,8,9

$$\text{Range} = 9 - 2 = 7$$

- Quartiles

- Quartiles are points taken at regular intervals of a data distribution.
- Used to divide the data into essentially equal size consecutive set

Measuring the dispersion of Data



- The first quartile (Q_1) is the value where 25 percentile of the data values are below it
- Second quartile (Q_2) is the value where 50 percentile of the data values are below it (same as the median)
- The third quartile (Q_3) is the value where 75 percentile of the data values are below it.
- 25% of the data is between the minimum and Q_1 , 25% is between Q_1 and the median, 25% is between the median and Q_3 , and 25% is between Q_3 and the maximum value.

The distance between the first and third quartiles is a simple measure of spread that gives the range

This distance is called the **interquartile range (IQR)** and is defined as $IQR = Q_3 - Q_1$.

Percentile Computation Procedure

- Organize the data into an ascending ordered array
- Calculate the p th percentile location:

$$i = \frac{P}{100}(n)$$

- Determine the percentile's location and its value.
- If i is a **whole number**, the percentile is the average of the values at the i and $(i+1)$ positions
- If i is **not a whole number**, the percentile is at the $(i+1)$ position in the ordered array

Percentile Example

- Raw Data: 14, 12, 19, 23, 5, 13, 28, 17
- Ordered Array: 5, 12, 13, 14, 17, 19, 23, 28
- Location of 30th percentile:

$$i = \frac{30}{100}(8) = 2.4$$

- The location index, i , is not a whole number; $i+1 = 2.4+1=3.4$; the whole number portion is 3; the 30th percentile is at the 3rd location of the array; the 30th percentile is 13.

Quartiles: Example

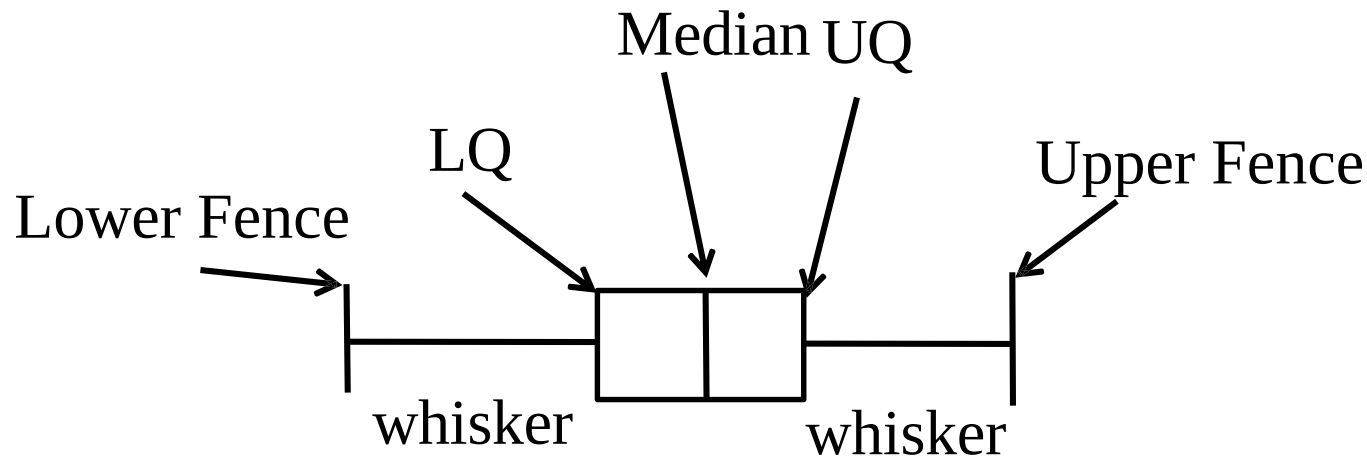
- Ordered array: 106, 109, 114, 116, 121, 122, 125, 129

- Q_1
$$i = \frac{25}{100}(8) = 2 \quad Q_1 = \frac{109 + 114}{2} = 111.5$$

- Q_2 :
$$i = \frac{50}{100}(8) = 4 \quad Q_2 = \frac{116 + 121}{2} = 118.5$$

- Q_3 :
$$i = \frac{75}{100}(8) = 6 \quad Q_3 = \frac{122 + 125}{2} = 123.5$$

Box-plot Diagram



Box plot diagram also termed as Whisker's plot is a graphical method typically depicted by quartiles and inter quartiles that helps in defining the upper limit and lower limit beyond which any data lying will be considered as outliers.

- Calculation of Inter Quartile Range (IQR)
 $= Q_3 - Q_1$
- The lower inner fence $= Q_1 - (1.5 * IQR)$
- The Upper inner fence $= Q_3 + (1.5 * IQR)$

Five number summary are : Min, Q1, Q2, Q3, Max

Outliers

- Outlier is a value that lies in a data series on its extremes, which is either very small or large and thus can affect the overall observation made from the data series
- Outliers are usually treated as abnormal values that can affect the overall observation due to its very high or low extreme values and hence should be discarded from the data series

Calculate Five Number Summary & Outliers (if any)

Let the data values are

(i) 199, 201, 236, 269, 271, 278, 283, 291, 301, 303, 341

Let the data values are

(ii) 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

Perform Box Plot analysis for the following Data

Data:

11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,
40,45,45,45,45,71,72,73,75

Measuring the dispersion of Data

- **Variance**

- It is the measure of the deviation of a variable from the mean

- **Standard Deviation**

- The standard deviation is also referred as Root Mean Square, the square root of the variance
 - A low standard deviation means that the data observations tend to be very close to the mean, whereas a high standard deviation indicates that the data are spread out over a large range of values.

	Population	Sample
Variance	$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$	$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$
Standard deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$	$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$

Variance and Standard Deviation

- Calculate the variance for the below values
3,4,4,5,5,5,6,6,6,7,7,8,9

x	$\bar{x}$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
3	5.8	-2.8	7.84
4	5.8	-1.8	3.24
4	5.8	-1.8	3.24
5	5.8	-0.8	0.64
5	5.8	-0.8	0.64
5	5.8	-0.8	0.64
6	5.8	0.2	0.04
6	5.8	0.2	0.04
6	5.8	0.2	0.04
7	5.8	1.2	1.44
7	5.8	1.2	1.44
8	5.8	2.2	4.84
9	5.8	3.2	10.24
			Sum = 176.88

Variance = $176.88 / 13 = 13.60$

Standard Deviation = 3.68

Basic statistical descriptions of data

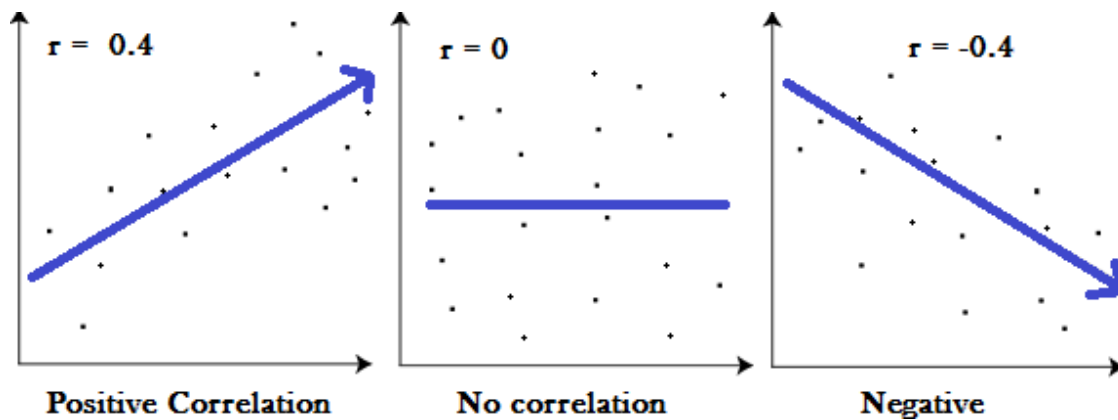
- **Covariance** – Measure to find out how much the dimensions vary from the mean with respect to each other (Two Dimensions)
- The formula for calculating covariance among two variables X and Y is

The **covariance** between A and B is defined as

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}.$$

Correlation coefficient for numeric data

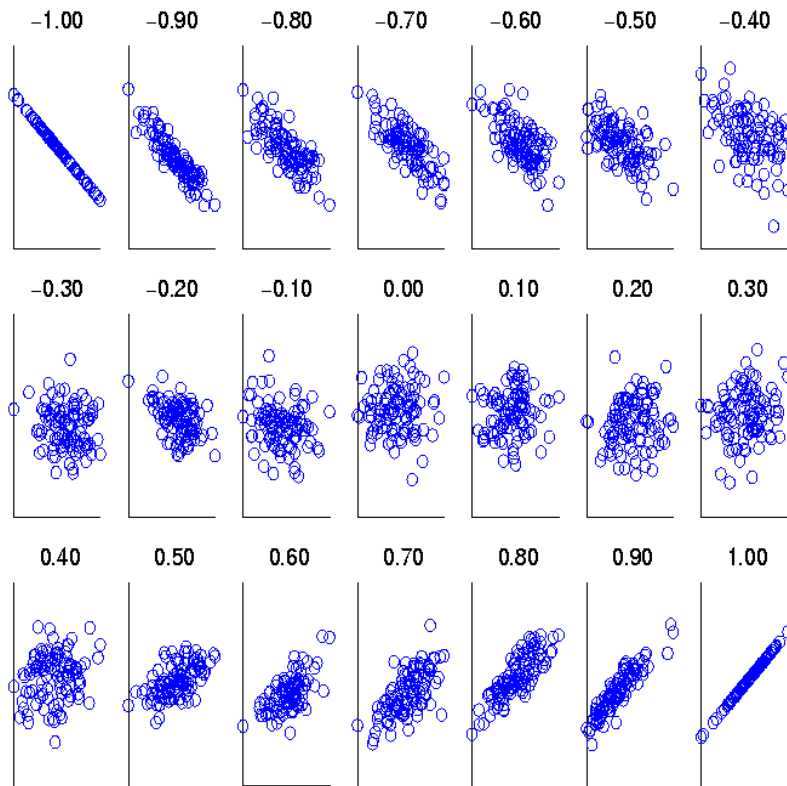
- Also known as **Pearson's product moment coefficient** . The formula return a value between -1 and +1
- If $r_{A,B}$ is greater than 0, then A and B are *positively correlated*, meaning that the values of A increase as the values of B increase. The higher the value, the stronger the correlation
- If the resulting value is equal to 0, then A and B are *independent*, and there is no correlation between them.



$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B}$$

Scatter plots can also be used to view correlations between attributes

Visualizing Changes of Correlation Coefficient



Correlation coefficient value range: $[-1, 1]$

A set of scatter plots shows sets of points and their correlation coefficients changing from -1 to 1

Correlation Analysis – Chi Square Test (Nominal Data)

- A hypothesis is an idea that can be tested.
- **Null Hypothesis** : There is no association between the two variables. Both are un-related with each other, Denoted H_0
- **Alternative hypothesis**: There is an association between the two variables, Denoted H_1
- Degree of Freedom & Level of Significance are the two terms used here
- The significance level, also denoted as alpha or ' α ', is the probability of rejecting the null hypothesis, usually 0.05
- Degrees of Freedom refers to the maximum number of logically independent values, which are values that have the freedom to vary,
$$DF = (r-1) \times (c-1)$$

Chi-Square Test : (For Nominal Data)

	Playing Cricket	Not playing Cricket	Sum (row)
Like science fiction	250	200	450
Not like science fiction	50	1000	1050
Sum(col.)	300	1200	1500

Chi Square Test (For Nominal Data)

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

- The Observed counts are those available in the data source.
- Expected counts are the projected frequencies in each cell if the null hypothesis is true (no association between the variables.)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

- If the critical value (found in the chi-square Table) is less than the calculated chi-square test value, then null hypothesis can be rejected

Chi-Square Table

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

Looking $df = 1$ and $\alpha = 0.05$, the critical value is 3.84.

Since 3.84 is less than the calculated chi-square value of 507.93, we reject the null hypothesis and state that there is a relationship between attributes “Play-cricket” & “like science fiction”

Do it Yourself

- 1) Find Mean, Median, Mode, Mid range, Range, Variance, Standard Deviation for the following Data :

12,23,34,44,59,70,98

- 2) Calculate Covariance, Correlation coefficient for the following dataset, and what the result indicates

Item Number:	1	2	3	4	5
x	10	39	19	23	28
y	43	13	32	21	20

Solve it

A group of students were classified in terms of personality trait (introvert or extrovert) and in terms of color preference (red, yellow, green or blue) with the purpose of seeing whether there is an association (relationship) between personality and colour preference. Data was collected from 400 students and presented in the 2 (rows) x 4 (cols) contingency table below:

Contingency table or confusion matrix

(Observed counts)	Colours				
	Red	Yellow	Green	Blue	Totals
Introvert personality	20	6	30	44	100
Extrovert personality	180	34	50	36	300
Totals	200	40	80	80	400

Formation of Hypothesis

- H_0 (null hypothesis):
Color preference is not associated with personality
- H_1 (alternate hypothesis):
Color preference is associated with personality
- Probability of rejecting null hypothesis is 5%
- The number of students expected in each cell of the table if the null hypothesis is true, is calculated.

Calculation

The expected numbers (under the null hypothesis) in each cell are equal to row total multiplied by column total divided by grand total.

Thus for the introvert/red cell the expected number is

$$\frac{100 \times 200}{400} = 50$$

Similarly the expected counts for
rest of the cells

(Expected counts)	Colours				
	Red	Yellow	Green	Blue	Totals
Introvert personality	50	10	20	20	100
Extrovert personality	150	30	60	60	300
Totals	200	40	80	80	400

Calculation of Chi-square values

To calculate the chi-squared statistic the value of

$$\frac{(\text{observed frequency} - \text{expected frequency})^2}{\text{expected frequency}}$$

Needs to be calculated for each cell in the table. For the introvert/red cell this is

$$\frac{(20 - 50)^2}{50} = 18.00$$

Chi-Square Test: red, yellow, green, blue

Expected counts are printed below observed counts

	red	yellow	green	blue	Total
1	20 50.00	6 10.00	30 20.00	44 <u>20.00</u>	100
2	180 150.00	34 30.00	50 60.00	36 <u>60.00</u>	300
Total	200	40	80	<u>80</u>	400

$$\begin{aligned} \text{Chi-Sq} = & 18.000 + \underline{1.600} + 5.000 + 28.800 + \\ & 6.000 + \underline{0.533} + 1.667 + 9.600 = 71.200 \end{aligned}$$

Degree of Freedom for this problem

$$\begin{aligned}df &= (r - 1) \times (c - 1) \\&= (2-1) \times (4-1) \\&= 3\end{aligned}$$

Looking of Chi-Square Distribution

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

Looking up the critical value for $df = 3$ and $\alpha = 0.05$, the critical value is 7.81.

Since 7.81 is less than the calculated chi-square value of 71.2, we reject the null hypothesis and state that there is a relationship between colour preference and personality.

Similarity, Dissimilarity, and Proximity

- **Similarity measure or similarity function**
 - A real-valued function that quantifies the similarity between two objects
 - Measure how two data objects are alike: The higher value, the more alike
 - Often falls in the range $[0,1]$: 0: no similarity; 1: completely similar
- **Dissimilarity (or distance) measure**
 - Numerical measure of how different two data objects are
 - In some sense, the inverse of similarity: The lower, the more alike
 - Minimum dissimilarity is often 0 (i.e., completely similar)
 - Range $[0, 1]$ or $[0, \infty)$, depending on the definition
- **Proximity** usually refers to either similarity or dissimilarity

Data Matrix and Dissimilarity Matrix

Data matrix

- A data matrix of n data points with l dimensions

Data Matrix

$$D = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1l} \\ x_{21} & x_{22} & \dots & x_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nl} \end{pmatrix}$$

Dissimilarity (distance) matrix

- n data points, but registers only the distance $d(i, j)$ (typically metric)
- Usually symmetric, thus a triangular matrix
- **Distance functions** are usually different for real, Boolean, categorical, ordinal, ratio, and vector variables
- Weights can be associated with different variables based on applications and data semantics

Dissimilarity Matrix

$$\begin{pmatrix} 0 & & & \\ d(2,1) & 0 & & \\ \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & 0 \end{pmatrix}$$

Example : Numerical Data

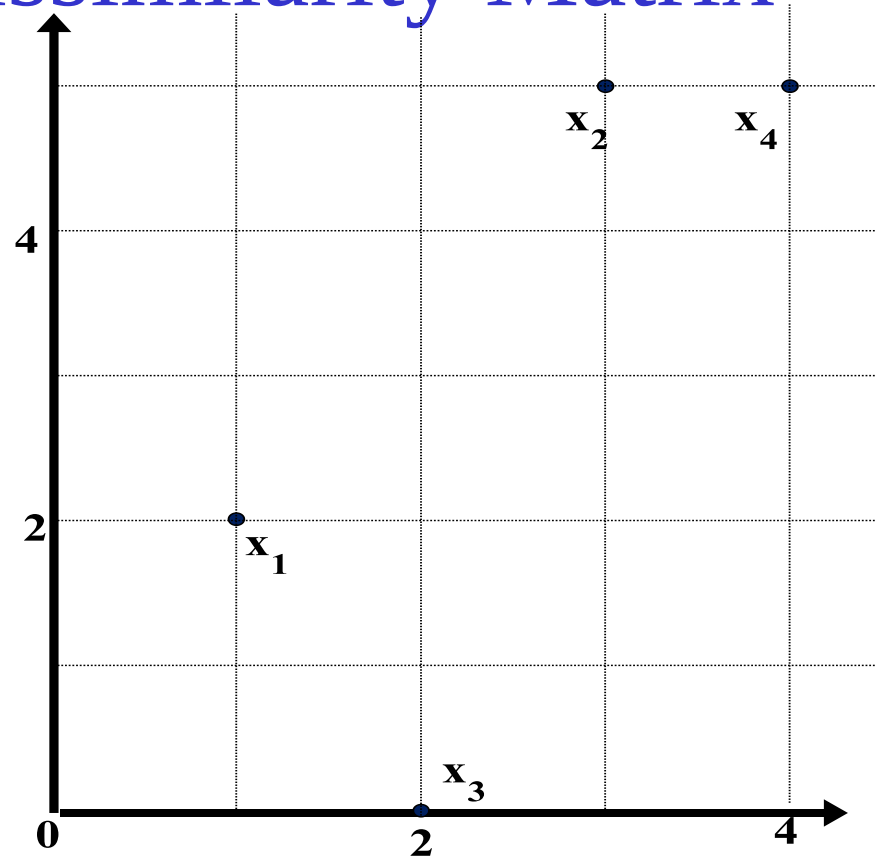
Data Matrix and Dissimilarity Matrix

Data Matrix

point	attribute1	attribute2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Dissimilarity Matrix
(by **Euclidean**
Distance)

	x_1	x_2	x_3	x_4
x_1	0			
x_2	3.61	0		
x_3	2.24	5.1	0	
x_4	4.24	1	5.39	0



Proximity Measure for Nominal Attributes

- How is dissimilarity computed between objects described by nominal attributes?
- The dissimilarity between two objects i and j can be computed based on the ratio of mismatches:

$$d(i, j) = \frac{p - m}{p},$$

where m is the number of matches (i and j are in the same state), and p is the total number of attributes describing the objects

Example

Object Identifier	Test-1 (nominal)
1	code A
2	code B
3	code C
4	code A

Dissimilarity Matrix

$$\begin{bmatrix} 0 & & & \\ d(2, 1) & 0 & & \\ d(3, 1) & d(3, 2) & 0 & \\ d(4, 1) & d(4, 2) & d(4, 3) & 0 \end{bmatrix}.$$

Dissimilarity Matrix

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

Proximity Measure for Nominal Attributes

Example:

Consider two objects

Object A: (Color: Red, Shape: Circle, Size: Large)

Object B: (Color: Blue, Shape: Square, Size: Large)

- **Identify the total number of attributes (p):** In this example, there are 3 attributes (Color, Shape, Size). So, $p = 3$.
- **Count the number of matches (m):**
 - Color: Red vs. Blue (Mismatch)
 - Shape: Circle vs. Square (Mismatch)
 - Size: Large vs. Large (Match)

Only the "Size" attribute matches. So, $m = 1$.

- **Calculate the dissimilarity ($d(A, B)$):**

The formula for dissimilarity is: $d(i, j) = (p - m)/p$

$$d(A, B) = (3 - 1)/3 = 2/3 \approx 0.67$$

This indicates that approximately 67% of the nominal attributes between Object A and Object B are dissimilar. A higher value indicates greater dissimilarity, while a value of 0 would indicate perfect similarity (all attributes match).

Proximity Measure for Binary Attributes

If all binary attributes are thought of as having the same weight, we have the 2×2 contingency table

Contingency table for binary attributes

		Object j		sum
		1	0	
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
sum		$q + s$	$r + t$	p

Where,

q is the number of attributes that equal 1 for both objects i and j

r is the number of attributes that equal 1 for object i but equal 0 for object j

s is the number of attributes that equal 0 for object i but equal 1 for object j

t is the number of attributes that equal 0 for both objects i and j .

The total number of attributes is p ,

where $p = q + r + s + t$.

Proximity Measure for Binary Attributes

Symmetric binary dissimilarity :

$$d(i, j) = \frac{r + s}{q + r + s + t}.$$

Asymmetric binary dissimilarity

$$d(i, j) = \frac{r + s}{q + r + s}.$$

Asymmetric binary similarity

Also called the **Jaccard coefficient**

$$sim(i, j) = \frac{q}{q + r + s}$$

Example : Asymmetric Binary Dissimilarity

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N



Gender is a symmetric attribute
(not counted in)

The remaining attributes are
asymmetric binary

Let the values Y and P be 1,
and the value N be 0

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	1	0	1	0	0	0
Mary	1	0	1	0	1	0
Jim	1	1	0	0	0	0

Example : Asymmetric Binary Dissimilarity

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	1	0	1	0	0	0
Mary	1	0	1	0	1	0
Jim	1	1	0	0	0	0

$$d(i, j) = \frac{r + s}{q + r + s}$$

$$d(\text{jack}, \text{mary}) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

Jack

Mary

	1	0	Σ_{row}
1	2	0	2
0	1	3	4
Σ_{col}	3	3	6

$$d(\text{jack}, \text{jim})$$

Calculate :

$$d(\text{jim}, \text{mary})$$

Which pair of objects are closer to each other ?

Distance on Numeric Data: Minkowski Distance

- **Minkowski distance**: A popular distance measure

$$d(i, j) = \sqrt[p]{|x_{i1} - x_{j1}|^p + |x_{i2} - x_{j2}|^p + \dots + |x_{il} - x_{jl}|^p}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{il})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jl})$ are two l -dimensional data objects, and p is the order (the distance so defined is also called L- p norm)

- Properties

$d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$ (Positivity)

$d(i, j) = d(j, i)$ (Symmetry)

$d(i, j) \leq d(i, k) + d(k, j)$ (Triangle Inequality)

- A distance that satisfies these properties is a **metric**

Special Cases of Minkowski Distance

$p = 1$: (L_1 norm) **Manhattan (or city block) distance**

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{il} - x_{jl}|$$

$p = 2$: (L_2 norm) **Euclidean distance**

$$d(i, j) = \sqrt{|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \cdots + |x_{il} - x_{jl}|^2}$$

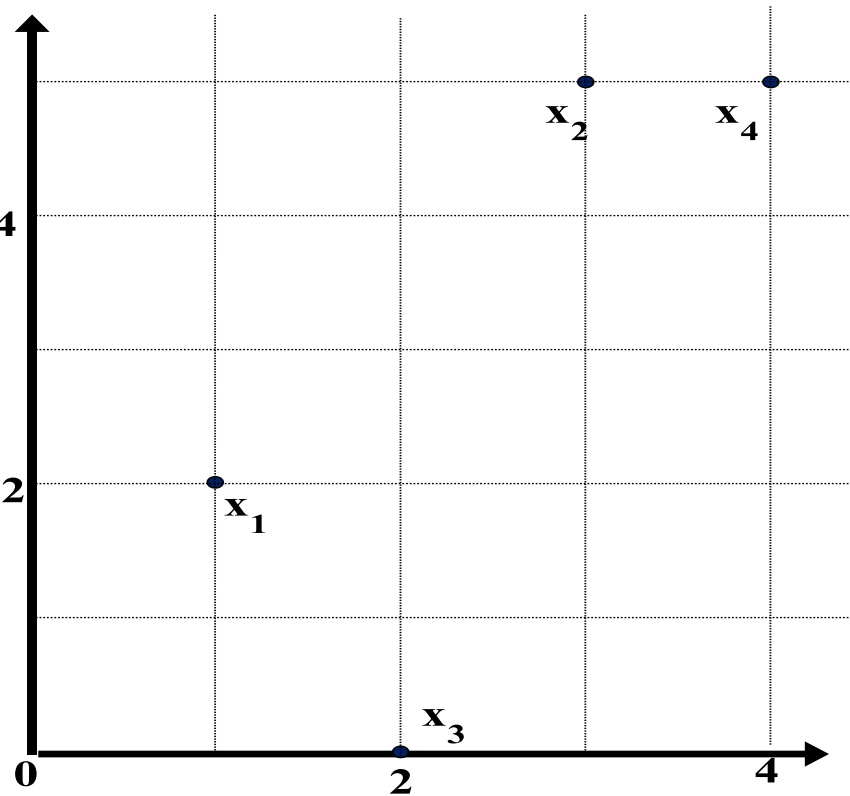
$p \rightarrow \infty$: ($L_{\max}$ norm, L_{∞} norm) **“supremum” distance**

The maximum difference between any component (attribute) of the vectors

$$d(i, j) = \lim_{p \rightarrow \infty} \sqrt[p]{|x_{i1} - x_{j1}|^p + |x_{i2} - x_{j2}|^p + \cdots + |x_{il} - x_{jl}|^p} = \max_{f=1}^l |x_{if} - x_{jf}|$$

Example : Distance on Numeric Data

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Supremum (L_∞)

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Proximity measure for Ordinal Variables

- An ordinal variable can be discrete or continuous
- Order is important, e.g., rank (e.g., Low, Medium, High)
- Can be treated like interval-scaled
 - Replace *an ordinal variable value* by its rank:
 - Map the range of each variable onto $[0, 1]$ by replacing i -th object in the f -th variable by

$$r_{if} \in \{1, \dots, M_f\}$$

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

- Compute the dissimilarity using methods for interval-scaled variables

Example : Proximity measure for Ordinal Variables

Object Identifier	Test
1	Excellent
2	Good
3	Poor
4	Good
5	Very Good
6	Fair
7	Poor
8	Good
9	Fair
10	Very Good

Step 1: Replace each value for the Test attribute by its rank

Object Identifier	Test
1	5
2	3
3	1
4	3
5	4
6	2
7	1
8	3
9	2
10	4

Example : Proximity measure for Ordinal Variables

Step 2: Normalize the ranking

Now that we have assigned ranks to each data point. Next, normalize these ranks so that they fall in the range of 0.0 to 1.0.

We can map ranks with the help of the following formula:

$$\text{Normalized Value} = \frac{X - 1}{M_f - 1}$$

M_f represents the total number of rank types or categories in the Test attribute, which is 5 for our case and X is the ordinal numeric value.

Excellent	For <i>Excellent</i> X=5	<i>Excellent</i> = $5 - 1/5 - 1 = 1.0$
Very Good	For <i>Very Good</i> X=4	<i>Very Good</i> = $4 - 1/5 - 1 = 0.75$
Good	For <i>Good</i> X=3	<i>Good</i> = $3 - 1/5 - 1 = 0.5$
Fair	For <i>Fair</i> X=2	<i>Fair</i> = $2 - 1/5 - 1 = 0.25$
Poor	For <i>Poor</i> X=1	<i>Poor</i> = $1 - 1/5 - 1 = 0$

Object Identifier	Test
1	1
2	0.5
3	0
4	0.5
5	0.75
6	0.25
7	0
8	0.5
9	0.25
10	0.75

Step 3: Use Euclidean distance to find the dissimilarity matrix

Mixed mode Attributes

How can we compute the dissimilarity between objects of mixed attribute types?"

- A more preferable approach is to process all attribute types together, performing a single analysis.
- One such technique combines the different attributes into a single dissimilarity matrix, bringing all of the meaningful attributes onto a common scale of the interval [0.0, 1.0].

Suppose that the data set contains p attributes of mixed types. The dissimilarity $d(i, j)$ between objects i and j is defined as

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}},$$

where the indicator $\delta_{ij}^{(f)} = 0$ if either (1) x_{if} or x_{jf} is missing (i.e., there is no measurement of attribute f for object i or object j), or (2) $x_{if} = x_{jf} = 0$ and attribute f is asymmetric binary; otherwise, $\delta_{ij}^{(f)} = 1$. The contribution of attribute f to the dissimilarity between i and j (i.e., $d_{ij}^{(f)}$) is computed dependent on its type:

- If f is numeric: $d_{ij}^{(f)} = \frac{|x_{if} - x_{jf}|}{\max_f - \min_f}$, where $\max_f$ and $\min_f$ are the maximum and minimum values of attribute f , respectively;
- If f is nominal or binary: $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$; otherwise, $d_{ij}^{(f)} = 1$; and
- If f is ordinal: compute the ranks r_{if} and $z_{if} = \frac{r_{if} - 1}{M_f - 1}$, and treat z_{if} as numeric.

Mixed mode Attributes

Example

Object Identifier	Test-1 (nominal)	Test-2 (ordinal)	Test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

Calculate $d(3,1)$

For Test -3 : $|64-45| / (64-22) = 19/42 = 0.45$

For Test -2 :

Fair : 1, good : 2, excellent : 3

Normalize it :

excellent : $(3-1) / (3-1) = 1/1 = 1$

fair : $(1-1) / (3-1) = 0$

good : $(2-1) / (3-1) = \frac{1}{2} = 0.5$

Test-2
1
0
0.5
1

 $|0.5-1| / (1-0) = 0.5/1 = 0.5$

For Test -1 :

Test 1 is the nominal feature,

Here $x_{if} \neq x_{jf}$, **code C** in the third instance, and **code A** in the first instance.

Hence the distance is 1.

Mixed mode Attributes

d(3,1)

$$d(3, 1) = \frac{1(1) + 1(0.50) + 1(0.45)}{3} = 0.65$$

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}},$$

Cosine Similarity of Two Vectors

- A **document** can be represented by a bag of terms or a long vector, with each attribute recording the *frequency* of a particular term (such as word, keyword, or phrase) in the document

Document	teamcoach		hockey	baseball	soccer	penalty	score	win	loss	season
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Other vector objects: Gene features in micro-arrays
- Applications: Information retrieval, biologic taxonomy, gene feature mapping, etc.
- Cosine measure: If d_1 and d_2 are two vectors (e.g., term-frequency vectors), then

$$\cos(d_1, d_2) = \frac{d_1 \bullet d_2}{\|d_1\| \times \|d_2\|}$$

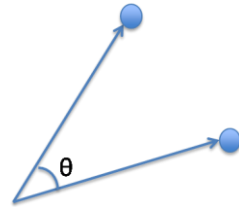
where $\bullet$ indicates vector dot product, $\|d\|$: the length of vector d

Example: Calculating Cosine Similarity

Calculating Cosine Similarity:

$$\cos(d_1, d_2) = \frac{d_1 \bullet d_2}{\|d_1\| \times \|d_2\|} \quad \text{sim}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

where $\bullet$ indicates vector dot product, $\|d\|$: the length of vector d



Ex: Find the **similarity** between documents 1 and 2.

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0) \quad d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

First, calculate vector dot product $d_1 \bullet d_2$

$$\begin{aligned} &= (5 \times 3) + (0 \times 0) + (3 \times 2) + (0 \times 0) + (2 \times 1) + (0 \times 1) + (0 \times 1) + (2 \times 1) + (0 \times 0) + (0 \times 1) \\ &= 25 \end{aligned}$$

Then, calculate $\|d_1\|$ and $\|d_2\|$

$$\|d_1\| = \sqrt{5 \times 5 + 0 \times 0 + 3 \times 3 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 0 \times 0} = 6.481$$

$$\|d_2\| = \sqrt{3 \times 3 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 1 \times 1 + 1 \times 1 + 0 \times 0 + 1 \times 1 + 0 \times 0 + 1 \times 1} = 4.12$$

Calculate cosine similarity: $\cos(d_1, d_2) = 25 / (6.481 \times 4.12) = 0.94$